

# Multi-Objective Path Selection Model and Algorithm for Emergency Evacuation<sup>\*</sup>

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**Abstract** - Evacuation planning is one of the most important aspects of emergency management. Path selection is one of the fundamental problems in evacuation planning. A multi-objective mathematical model is presented for path selection in emergency evacuation. The two objectives of the model are to minimize the total travel time along a path and to minimize the path complexity respectively. Taking into account the real-time effect of disaster extension, the travel speed on each arc is modeled as a continuous decrease function with respect to time. The complexity of the path is modeled as the total number of arcs included in the path. An ant colony optimization algorithm is proposed to solve the model. Simulation results show the effectiveness and feasibility of the model and algorithm.

**Index Terms** - Emergency evacuation, path selection, disaster extension, multi-objective model, ant colony optimization algorithm.

## I. INTRODUCTION

In recent years, frequent natural disasters and man-made catastrophic events have brought great loss to human beings [1, 2]. As an emerging research area, emergency management is attracting more and more attention of researchers [3, 4]. Evacuation planning is one of the most important aspects of emergency management. Residents must be evacuated to safety areas effectively under natural or man-made disasters such as flood, hurricane, fire and toxic leakage [5, 6]. There have been some researches on the evacuation planning problem and a series of network-based models and algorithms have been proposed. Reference [7] gave a very good review on the mathematical modelling of evacuation problems. Reference [8] studied the evacuation planning problem taking into account the capacity constraints of the arcs in the evacuation network. Reference [9] studied the traffic assignments problem in evacuation modelling.

Most of the above research works considered the travel time on each arc of the evacuation network as a constant. But related researches have shown that quite a few kinds of disasters will extend gradually as time goes on. For example, we can observe the diffusion of gas and smoke in a fire or toxic leakage [10, 11] and also the extension of hurricane disasters [12, 13]. These disasters all have the above characteristics when extending. So the travel conditions on the

arcs of the evacuation network may deteriorate continuously due to the extension of disasters. We can suppose that the travelling speed on each arc will decrease continuously due to real-time effect of disaster extension, and the decrease extent will differ with the positions of the arcs.

On the other hand, most of the existing research works on evacuation modelling took time as the most important parameter to be considered. The objective of these models was to minimize the time needed to complete the evacuation process. But the residents will be in great panic under dangerous conditions during the evacuation process and the sense of panic will affect the ability of the residents to follow a suitable path [14, 15]. Therefore the evacuation path should be as simple as possible so that the residents can follow the evacuation command more easily. The complexity of the evacuation path should also be taken into account as an objective of the evacuation path selection model.

In this study, we present a multi-objective path selection model for emergency evacuation. The two objectives of the model are to minimize total travel time along the path and to minimize the complexity of the path respectively. The travel speed on each arc is modelled as a continuous decrease function with respect to time taking into account the real-time effect of disaster extension. The complexity of the path is modelled as the total number of arcs included in a selected path. An ant colony optimization algorithm is proposed to solve the multi-objective optimization model. Simulation results show the effectiveness and feasibility of the model and algorithm.

## II. MULTI-OBJECTIVE PATH SELECTION MODEL FOR EMERGENCY EVACUATION

### A. Definition of Variables and Parameters

1) An emergency evacuation network is defined as a directed graph  $G(V, A)$  where  $V = \{v_1, v_2, \dots, v_n\}$  is the set of nodes and  $A \subseteq V \times V$  is the set of arcs. Let  $v_1, v_2, \dots, v_n$  denote the nodes in the evacuation network, where  $v_1$  is the source node and  $v_n$  is the destination node of the evacuation process.

2)  $l_{ij}$  denotes the length of the arcs between nodes  $v_i$  and  $v_j$ , where  $(v_i, v_j) \in A$ .

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3)  $s_{ij}^0$  is the traveling speed on arc  $(v_i, v_j)$  under normal conditions. Define  $s_{ij}(t)$  as the travel speed on arc  $(v_i, v_j)$  at time  $t$  under disaster conditions.

Observing the extension processes of some disasters such as flood and hurricane, we can find that the traveling speed on each arc of the evacuation network will decrease gradually with the extension of disasters as time goes on. The decrease extent of the traveling speed is affected by the position of the arc and the type of the disaster etc. Hence, without loss of generality, we assume the decrease function of the traveling speed as follows:

$$s_{ij}(t) = s_{ij}^0 \cdot \alpha_{ij} \cdot e^{-\beta_{ij}t}$$

Here,  $\alpha_{ij}, \beta_{ij}$  are the decrease parameters that determine the decrease extent of the travel speed function  $s_{ij}(t)$ .  $\alpha_{ij}, \beta_{ij}$  can be estimated according to the distance from arc  $(v_i, v_j)$  to the disaster center, the vulnerability of the arc  $(v_i, v_j)$  and the type of the disaster etc.

4) Let  $t_{ij}$  be the time needed to travel through arc  $(v_i, v_j)$ .  $t_i$  denotes the time when the residents reach node  $v_i$ ,  $t_j$  denotes the time when the residents reach node  $v_j$  along arc  $(v_i, v_j)$ . It is obvious that  $t_{ij} = t_j - t_i$ .

5)  $x_{ij}$  is the decision variable in the model.  $x_{ij}=1$  when arc  $(v_i, v_j)$  is included in the fixed path and  $x_{ij}=0$  when arc  $(v_i, v_j)$  is not included in the fixed path.

6)  $P$  denotes the evacuation path which is composed of a sequence of nodes in the network. Let  $p_k$  be the sequence number of node  $v_{p_k}$  in the network, then path  $P$  can be represented as  $(v_{p_1}, v_{p_2}, \dots, v_{p_k}, \dots, v_{p_K})$ , where  $1 \leq p_k \leq n$ ,  $k$  is the travel sequence of node  $v_{p_k}$  along path  $P$ . Path  $P$  starts at the source node and terminates at the destination node, that is,  $p_1 = 1$  and  $p_k = n$ . Path  $P$  should not have circles considering the time pressures in emergency evacuation.

7) Let  $ET(P, v_{p_k})$  denote the travel time from node  $v_{p_1}$  to node  $v_{p_k}$  along path  $P = (v_{p_1}, v_{p_2}, \dots, v_{p_k})$ , where  $1 \leq p_k \leq n$ . We can easily obtain:

$$ET(P, v_{p_k}) = \sum_{m=1}^{k-1} t_{p_m p_{m+1}} = (t_{p_2} - t_{p_1}) + (t_{p_3} - t_{p_2}) + \dots + (t_{p_k} - t_{p_{k-1}}) = t_{p_k}.$$

### B. Recursive Method for Computing the Travel Time of a Path

Based on the above definitions, for an already known path  $P = (v_{p_1}, v_{p_2}, \dots, v_{p_k})$ , we can obtain:

$$t_{p_1} = t_1 = 0 \quad (1)$$

$$\int_{p_{k-1}}^{p_k} s_{p_{k-1}p_k}(t)dt = l_{p_{k-1}p_k} \quad 2 \leq k \leq K \quad (2)$$

In (2), if we have already known the lower limit of the integral  $t_{p_{k-1}}$ , the integrand  $s_{p_{k-1}p_k}(t)$  and the integral result  $l_{p_{k-1}p_k}$ , the upper limit of the integral  $t_{p_k}$  can be obtained through solving (2).

Under real-time effect of disaster extension, the traveling speed on arc  $(v_i, v_j)$  is not a constant but a continuous decrease function with respect to time. From (2) we can see that the travel time  $t_{ij}$  on arc  $(v_i, v_j)$  is determined not only by the length of the arc  $l_{ij}$  and the travel speed function  $s_{ij}(t)$ , but it is also determined by the time when the residents reach the origin node  $v_i$  of arc  $(v_i, v_j)$ . The recurrence relations presented by (1) and (2) show that the total travel time of path  $P$  can be obtained through calculating the arriving time  $t_{p_k}$  at node  $v_{p_k}$  ( $1 \leq p_k \leq n$ ) recursively.

### C. The Two Objectives Of the Path Selection Model

In emergency evacuation, the evacuation time is the most important factor to be considered [7]. The residents must remove from the dangerous area to safety as quickly as possible. The first objective of the path selection model is to minimize total travel time along an evacuation path which can be represented as follows:

$$\min f_1 = \sum_{i=1}^n \sum_{j=1}^n t_{ij} x_{ij} \quad (3)$$

As described in part B of this section, the travel time along a path can be obtained through calculating the arriving time  $t_{p_k}$  at node  $v_{p_k}$  ( $1 \leq p_k \leq n$ ) recursively since the travel speed on each arc of the evacuation network is not a constant due to real-time effect of disaster extension.

Related researches have shown that most congestion and panic happen at the intersections of two arcs in the evacuation network. For example, movements through an exit are more complicated than that at free horizontal surfaces in the building evacuation [15] and most traffic delays in regional evacuations occur at intersections [16]. When traveling along a path, the fewer the arcs included in the path, the more easily the residents can follow an evacuation command. The complexity of an evacuation path can be simply represented as the total number of arcs included in a fixed path. So the second objective of the model is represented as follows:

$$\min f_2 = \sum_{i=1}^n \sum_{j=1}^n x_{ij} \quad (4)$$

### D. Model Formulation

Based on the above definitions and analysis, the formulation of multi-objective path selection model for emergency evacuation can be described as follows:

$$\min f_1 = \sum_{i=1}^n \sum_{j=1}^n t_{ij} x_{ij} \quad (3)$$

$$\min f_2 = \sum_{i=1}^n \sum_{j=1}^n x_{ij} \quad (4)$$

s.t.

$$\int_{t_i}^{t_j} s_{ij}(t) dt = l_{ij} \quad (5)$$

$$t_{ij} = t_j - t_i \quad (6)$$

$$t_1 = 0 \quad (7)$$

$$s_{ij}(t) = s_{ij}^0 \cdot \alpha_{ij} \cdot e^{-\beta_{ij}t} \quad (8)$$

$$\sum_{j=1, j \neq i}^n x_{ij} - \sum_{j=1, j \neq i}^n x_{ji} = \begin{cases} 1 & i = 1 \\ -1 & i = n \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

$$\sum_{j=1, j \neq i}^n x_{ij} \begin{cases} \leq 1 & i \neq n \\ = 0 & i = n \end{cases} \quad (10)$$

$$x_{ij} = 0, 1, \quad i = 1, 2, \dots, n; j = 1, 2, \dots, n \quad (11)$$

Where (5), (6) and (7) are the recursion formula of the total travel time along a path, which means that arc  $(v_i, v_j)$  is traveled through with the speed  $s_{ij}(t)$  during time period  $t_{ij}$ . Equation (8) is the decrease function of the traveling speed on arc  $(v_i, v_j)$  under disaster conditions. Constraint (9) restricts the value of  $x_{ij}$  to constitute a feasible path from the source node  $v_1$  to the destination node  $v_n$ . Constraint (10) ensures that there are no circles in the path. Constraint (11) is the 0-1 integer constraint of the decision variable  $x_{ij}$ .

### III. ANT COLONY OPTIMIZATION ALGORITHM FOR THE MODEL

The model presented in this paper is a multi-objective model and the traveling speed on each arc is not a constant but a continuous decrease function with respect to time. Therefore classical algorithms to solve shortest path problem can't be used to solve the presented model. As a novel metaheuristic algorithm, ant colony optimization (ACO) algorithm was proposed by Dorigo et al firstly to solve TSP problem [17]. In recent years, ACO has been applied to solve many different combinatorial problems successfully. ACO should also be suitable to solve the path selection problem presented in this paper since it was proposed based on the simulation of the way real ant find the shortest route between a food source and their nest. After converting the multi-objective path selection model into a single-objective model by ideal point method, an ACO algorithm is designed to achieve the weak efficient solution of the multi-objective path selection model for emergency evacuation.

#### A. Conversion to Single-Objective Model

First we use the ideal point method to deal with the model and convert the multi-objective path selection model into a single-objective model as follows:

$$\text{Min } F = r_1 \frac{f_1 - f_1^*}{f_1^*} + r_2 \frac{f_2 - f_2^*}{f_2^*} \quad (12)$$

s.t.

$$(5) - (11)$$

Here  $\bar{f}^* = (f_1^*, f_2^*)^T$  is the ideal point.

$\bar{r} \in R = \{\bar{r} | r_i \geq 0, \sum r_i = 1, i = 1, 2\}$  is the weight vector.

According to the theorem about ideal point method, the optimum solution of the model (12), (5)-(11) is the weak efficient solution of the model (3)-(11).

#### B. ACO Algorithm to Solve Single-Objective Path Selection Model

Here we design an ACO algorithm to solve single-objective path selection model named as SOPS-ACO. That is, SOPS-ACO is to find an optimal path from the source node to the destination node of the evacuation network considering only one objective function. The algorithm can be described as follows:

**Step1:** Initialize (cycles counter  $NC=0$ ). Set the value of fundamental parameters of the ACO algorithm including the maximum number of cycles  $NC_{\max}$ , the number of ants  $M$ . For every edge  $(v_i, v_j)$  set an initial pheromone value  $\tau_{ij}(0) = c$  and set the initial pheromone updating value  $\Delta\tau_{ij} = 0$ .

**Step2:** Every ant constructs a path from the source node to the destination node.

First place all the  $M$  ants on the source node. At node  $i$ , the  $m$ th ant chooses node  $j$  to move to with the probability:

$$p_{ij}^m = \begin{cases} \frac{\tau_{ij}^\alpha \cdot \eta_{ij}^\beta}{\sum_{j=1}^n \tau_{ij}^\alpha \cdot \eta_{ij}^\beta} & \text{if } (v_i, v_j) \in A \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

Here  $\eta_{ij}$  is a domain dependent heuristic parameter, the value of  $\eta_{ij}$  differs with different single-objective problems.  $\alpha, \beta$  are parameters that weight the relative importance of pheromone trail and heuristic information.

The process of the movement of the  $m$ th ant from one node to the next one continues until the destination node is included in the path constructed by the ant. And step2 finishes after all the  $M$  ants have constructed a path from the source node to the destination node.

**Step3:** Pheromone updating. After all the  $M$  ants have finished their movement from the source node to the destination node in one cycle, the pheromone on each arc is updated with (14)-(16):

$$\Delta\tau_{ij}^m = \begin{cases} Q/L_m, & \text{if the } m\text{th ant passed arc } (v_i, v_j) \text{ in this cycle} \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

$$\Delta\tau_{ij} = \sum_{m=1}^M \Delta\tau_{ij}^m \quad (15)$$

$$\tau_{ij}(NC+1) = (1-\rho) \cdot \tau_{ij}(NC) + \Delta\tau_{ij} \quad (16)$$

Here  $L_m$  is a value related to the value of the objective function.  $\rho$  is the evaporation rate of the pheromone.

Step4: If  $NC < NC_{max}$ ,  $NC + 1 \rightarrow NC$ , go to step2; else the algorithm is terminated.

#### C. Algorithm to Solve the Multi-Objective Path Selection Model

To solve the multi-objective path selection model presented in this paper, firstly we can get the ideal point of the multi-objective problem. And then we can put the ideal point into (12) and solve the single-objective model to get the weak efficient solution of the original multi-objective model. So the weak efficient solution of the multi-objective path selection model can be achieved by solving three single-objective models. By using suitable heuristic values and pheromone updating methods, SOPS-ACO algorithm can be applied to solve all the three single-objective models effectively. The algorithm to solve the multi-objective path selection model is as follows:

**Step1:** Use SOPS-ACO to get the optimal solution of the single-objective model (3), (5)-(11) and obtain the corresponding optimal value  $f_1^*$ .

In the traveling speed decrease function (8), bigger  $s_{ij}^0 \cdot \alpha_{ij} \cdot e^{-\beta_{ij}}$  means lower decrease extent of the speed on arc  $(v_i, v_j)$ , so the heuristic value used here is as follows:

$$\eta_{ij}^1 = \begin{cases} s_{ij}^0 \cdot \alpha_{ij} \cdot e^{-\beta_{ij}} & \text{if } (v_i, v_j) \in A \\ 0 & \text{otherwise} \end{cases} \quad (17)$$

In this step,  $L_m$  in the pheromone updating function (14) is equal to the travel time along the path found by ant  $m$  which can be calculated by the recursive method described in part B of section II.

**Step2:** Use SOPS-ACO to get the optimal solution of the single-objective model (4), (5)-(11) and obtain the corresponding optimal value  $f_2^*$ .

In objective function (4), there is no difference among the weights of the arcs, so the heuristic value used here is as follows:

$$\eta_{ij}^2 = \begin{cases} 1 & \text{if } (v_i, v_j) \in A \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

Here  $L_m$  in the pheromone updating function (14) is equal to the number of arcs included in the path found by ant  $m$  which can be easily obtained.

**Step3:** Put  $f_1^*$  and  $f_2^*$  into objective function (12) and apply SOPS-ACO to get the optimal solution of model (12), (5)-(11).

The heuristic value used in this step is the same as (17) for the heuristic value in step2 has no difference on different arcs.  $L_m$  in this step is equal to the value of objective function (12) corresponding to the path found by ant  $m$ , that is,

$$L_m = r_1 \frac{f_1^m - f_1^*}{f_1^*} + r_2 \frac{f_2^m - f_2^*}{f_2^*} \quad (19)$$

Here  $f_1^m$  and  $f_2^m$  are the values of objective function (3) and (4) corresponding to the path found by ant  $m$  respectively. And  $f_1^*$ ,  $f_2^*$  are the optimal value of objective function (3) and (4) obtained in step1 and step2. The solution obtained in step3 is a weak efficient solution of the multi-objective path selection model.

#### IV. NUMERICAL EXAMPLE

We apply the model and algorithm to an evacuation network with 20 nodes. The structure of the network is shown in Fig. 1. Suppose the disaster happens at node 1. The length of each arc  $l_{ij}$ , the initial traveling speed  $s_{ij}^0$  and the decrease parameters  $\alpha_{ij}$ ,  $\beta_{ij}$  are shown in Table 1.

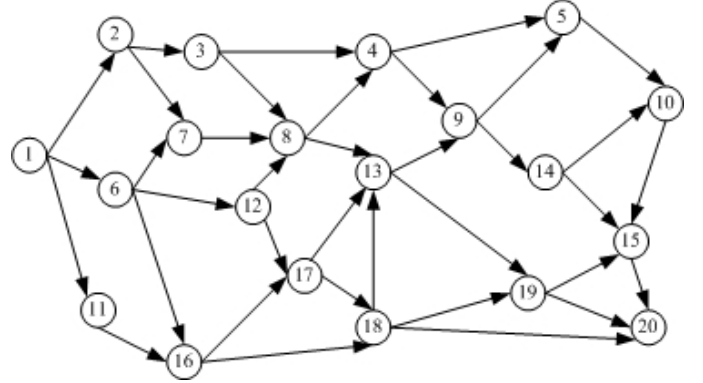


Fig. 1 The structure of the emergency evacuation network

The ACO algorithm was implemented in C and run on a PC with 2.66GHz CPU, 512MB RAM. Parameters of ACO are set as: the number of ants  $M=10$ , the maximum number of cycles  $NC_{max}=50$ , the initial pheromone value  $c=1000$ ,  $\alpha = 2$ ,  $\beta=1$ ,  $Q=1$ ,  $\rho=0.2$ . When  $r_1=0.9, r_2=0.1$ , the ideal point and the weak efficient solution to the multi-objective model are shown in Table 2. From the results shown in Table 2 we can see that the weak efficient solution is a trade-off between the solutions corresponding to the ideal point.

TABLE I  
PARAMETERS OF THE EMERGENCY EVACUATION NETWORK

| $(v_i, v_j)$ | $(l_{ij}, s_{ij}^0, \alpha_{ij}, \beta_{ij})$<br>(m, m/min, -, -) | $(v_i, v_j)$ | $(l_{ij}, s_{ij}^0, \alpha_{ij}, \beta_{ij})$<br>(m, m/min, -, -) | $(v_i, v_j)$ | $(l_{ij}, s_{ij}^0, \alpha_{ij}, \beta_{ij})$<br>(m, m/min, -, -) |
|--------------|-------------------------------------------------------------------|--------------|-------------------------------------------------------------------|--------------|-------------------------------------------------------------------|
| (1,2)        | (50,100,0.5,0.4)                                                  | (6,16)       | (100,100,0.4,0.3)                                                 | (14,10)      | (80,100,0.95,0.1)                                                 |
| (1,6)        | (20,70,0.2,0.5)                                                   | (7,8)        | (40,80,0.5,0.2)                                                   | (14,15)      | (60,80,0.9,0.08)                                                  |
| (1,11)       | (70,90,0.6,0.2)                                                   | (8,4)        | (40,90,0.6,0.15)                                                  | (15,20)      | (30,80,1,0.05)                                                    |
| (2,3)        | (20,70,0.5,0.3)                                                   | (8,13)       | (30,100,0.5,0.2)                                                  | (16,17)      | (80,90,0.6,0.3)                                                   |
| (2,7)        | (40,80,0.5,0.25)                                                  | (9,5)        | (70,100,0.8,0.15)                                                 | (16,18)      | (110,100,0.6,0.25)                                                |
| (3,4)        | (80,90,0.55,0.2)                                                  | (9,14)       | (40,80,0.7,0.2)                                                   | (17,13)      | (40,80,0.5,0.2)                                                   |
| (3,8)        | (60,90,0.6,0.2)                                                   | (10,15)      | (50,100,0.9,0.08)                                                 | (17,18)      | (40,80,0.7,0.15)                                                  |
| (4,5)        | (110,80,0.8,0.15)                                                 | (11,16)      | (30,80,0.5,0.25)                                                  | (18,13)      | (60,90,0.6,0.15)                                                  |

|        |                  |         |                   |         |                    |
|--------|------------------|---------|-------------------|---------|--------------------|
| (4,9)  | (40,100,0.9,0.3) | (12,8)  | (20,80,0.4,0.4)   | (18,19) | (70,100,0.8,0.1)   |
| (5,10) | (60,100,1,0.1)   | (12,17) | (40,90,0.7,0.2)   | (18,20) | (130,80,0.95,0.08) |
| (6,7)  | (30,90,0.3,0.3)  | (13,9)  | (30,100,0.6,0.2)  | (19,15) | (40,80,0.8,0.05)   |
| (6,12) | (60,80,0.5,0.2)  | (13,19) | (110,90,0.8,0.15) | (19,20) | (70,100,0.9,0.08)  |

TABLE II  
IDEAL POINTS AND WEAK EFFICIENT SOLUTION OBTAINED BY ACO WHEN  $r_1 = 0.9, r_2 = 0.1$

|                         | The selected path                 | $f_1$ (The travel time along the path) | $f_2$ (Complexity of the path) | $F$ (Fitness value of the weak efficient solution) |
|-------------------------|-----------------------------------|----------------------------------------|--------------------------------|----------------------------------------------------|
| Ideal point $f_1^*$     | 1 → 2 → 3 → 8 → 13 → 19 → 15 → 20 | 13.597900s                             | 7                              | --                                                 |
| Ideal point $f_2^*$     | 1 → 11 → 16 → 18 → 20             | 18.126943s                             | 4                              | --                                                 |
| Weak efficient solution | 1 → 2 → 3 → 8 → 13 → 19 → 20      | 13.877971s                             | 6                              | 0.068537                                           |

## V. CONCLUSIONS

In this study, a multi-objective path selection model is built for emergency evacuation. The two objectives of the model are to minimize total travel time along the path and to minimize the complexity of the path respectively. Since the model is a multi-objective one and the traveling speed on each arc is modeled as a continuous decrease function with respect to time, classical algorithms for shortest path problem can't be used to solve the model proposed in this paper. An ant colony optimization algorithm is proposed to solve the multi-objective optimization model. Simulation results on small scale evacuation network show the effectiveness and feasibility of the model and algorithm. To test the performance of the algorithm on large scale evacuation networks and design improved algorithms to solve large scale problems efficiently will be one of the future research directions of the authors.

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